3. Requirements Analysis

Functional requirements

ID | Requirement | Acceptance evidence

FR-01 | Read the supplied CSV and record row/column counts | Verified profile

FR-02 | Describe all {df.shape[1]} fields and units where known | Data dictionary

FR-03 | Report missing values, duplicates, and suspicious ranges | Quality audit

FR-04 | Create category, geography, model-year, and metric comparisons | Tableau worksheets

FR-05 | Define all KPIs and calculations | Calculation register

FR-06 | Provide interactive filters where they add analytical value | Tested dashboard

FR-07 | Publish Tableau dashboard and story | Public view URLs

FR-08 | Embed both views in website | Responsive HTML page

FR-09 | Provide direct-view fallbacks | Visible links

FR-10 | Document results and limitations | Final report

Non-functional requirements

- **Accuracy:** numeric summaries must reconcile with the source or documented cleaning rules.
- **Clarity:** charts must identify the metric, unit, and aggregation.
- **Responsiveness:** page and embeds must be usable on desktop, tablet, and mobile.
- **Accessibility:** headings, contrast, and link text should be understandable.
- **Reliability:** direct Tableau links should remain available if an embed fails.
- **Traceability:** conclusions must cite the metric and filter context supporting them.
- **Honesty:** unknown provenance and field definitions must be disclosed.

Acceptance criteria

The file has 100,000 rows and 37 columns; `vehicle_id` has 100,000 distinct values; complete duplicate rows and duplicate IDs are absent in the raw file; missingness is documented; every KPI has a definition; and the final website loads both views or displays a useful fallback link.

Stakeholders

- Project author/team: analysis, Tableau build, web integration.
- Instructor/reviewer: evaluate method, clarity, and evidence.
- Website user: explore views and understand scope.